Non-Lattice Simulations of Supersymmetric Anharmonic Oscillator using Complex Langevin Method

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Abstract

Lattice simulations of supersymmetric field theories are not straightforward. In some cases they demand complicated fine tuning of coupling constants to obtain the desired supersymmetric point in the continuum limit. The chiral and/or Majorana nature of fermions makes the situation even worse and it becomes a cumbersome task to even formulate an appropriate lattice theory.

In this report we use a new method of simulation (*non-lattice method*), which is a more effective and elegant method to simulate supersymmetric theories. We provided the simulation results for the supersymmetric anharmonic oscillator. We extracted bosonic mass gaps from the exponential decay of the two-point correlators. We have used complex Langevin method for simulations and also discussed its dynamics in great detail.

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Contents

1 Introduction

The algebra of supersymmetry contains generators of continuous translations. On a lattice continuous translations are replaced by discrete ones and thus supersymmetry is broken on the lattice. One has to do cumbersome fine tuning of the coupling constants in order to recover supersymmetry as continuum limit of the lattice theory is taken. The aim of this report is to point out that there exists a very simple and efficient method to simulate supersymmetric gauge theories in one dimension.

This report is organized as follows. In Sec. 2 we discuss the basic framework of complex Langevin method (CLM) and how to use it for taming the sign problem. In Sec. 3 non-lattice method is discussed. In Sec. 4 we have provided some preliminary complex Langevin simulation results. In Sec. 5 we motivated some future research and concluded the report.

2 Complex Langevin Dynamics

2.1 Sign Problem

In Euclidean field theory, the expectation value of a certain observable $O(\phi)$ is given by

$$\langle O(\phi)\rangle = \frac{\int [D\phi]\ O(\phi)e^{-S(\phi)}}{Z}, \tag{2.1}$$

$$Z = \int [D\phi]\ e^{-S(\phi)}, \tag{2.2}$$

where ϕ is a field, and $S[\phi]$ is the Euclidean action of the system, which is normally real.

We can apply the standard Monte Carlo methods, Metropolis or heat bath algorithm for the numerical simulation of the quantity in Eq. (2.1). But field theories having complex actions are difficult to treat non-perturbatively since the weight $e^{-S} = \left|e^{-S}\right| e^{i\phi}$ in the partition function is not real. Standard numerical approaches based on a probability interpretation and importance sampling like Monte Carlo method will then typically break down, which is commonly referred to as the *sign problem*. But, one often encounters systems that have complex actions like gauge theories with fermions, field theories in Minkowski space, QCD systems with finite chemical potentials at finite temperature etc. There are a few methods available to tame this sign problem like analytical continuation, reweighting, tensor networks, etc. These methods also have their own limitations. Apart from these, we have numerical methods. Mostly we sort to these numerical methods because they have shown promising results in evading the sign problem.

2.1.1 Re-weighting Procedure

A field theory with a non-positive weight can be transformed into one with positive weight by incorporating the non-positive part (sign or complex phase) of the weight into the observable. In principle, one can calculate the expectation of an observable $O(\phi)$, i.e., with $S = S_R + iS_I$ as

$$\langle O(\phi)\rangle_S = \frac{\langle O(\phi)e^{-iS_I(\phi)}\rangle_{S_R}}{\langle e^{-iS_I(\phi)}\rangle_{S_R}}, \tag{2.3}$$

where subscript S_R in Eq. (2.3) indicates path integral with weight e^{-S_R}.

Now the desired expectation value is a ratio where the numerator and denominator are expectation values that both use a positive weighting function. However, the phase e^{-iS_I} is a highly oscillatory function in the configuration space, so if one uses Monte Carlo methods to evaluate the numerator and denominator, each of them will evaluate to a very small number, whose exact value is swamped by the noise inherent in the Monte Carlo sampling process. In such case, configurations generated by the probability distribution e^{-S_R} are irrelevant to the whole system. This occurs for large volumes or for field theories having strong coupling. Therefore, we need to find a method by which the configurations distributed according to a *complex probability distribution* $\sim e^{-S}$ are directly generated. A possible algorithm which helps us in realizing complex probability distribution is Langevin algorithm.

2.2 Langevin Algorithm

Numerical simulations of lattice field theories with a complex actions are not possible with standard Monte Carlo methods in the case where the path integral weight is not positive-definite. Complex Langevin (CL) dynamics provides an appealing alternative since it does not rely on importance sampling.

Complex Langevin equations were proposed by Parisi [1] and Klauder [2] to simulate systems with complex valued path integral weights. For d dimensional systems with real weights, a Langevin equation in $d+1$ dimensions may be used to study the partition function of the system.

In complex Langevin method, we stochastically evolve our field configurations in fictitious time t. The evolution is governed by Langevin equation

$$\frac{d\phi}{dt} = -\frac{\delta S(\phi)}{\delta \phi} + \eta(t), \tag{2.4}$$

where $S'(\phi) = -\frac{\delta S(\phi)}{\delta \phi}$ is the drift term coming from real action functional, t is fictitious time and η is Gaussian white noise. The noise $\eta(t)$ satisfy the following properties

$$\langle \eta(t) \rangle_\eta = 0; \qquad \langle \eta(t)\eta(t') \rangle_\eta = 2\delta(t - t'). \tag{2.5}$$

The long time average of the quantity O over the solution of the above Langevin equation is supposed to give us the expectation value in Eq. (2.1).

$$\frac{1}{T} \int_{t_{\rm th}}^{t_{\rm th}+T} dt \; O(\phi_n(t)) \quad \xrightarrow[T,\, t_{\rm th} \to \infty]{} \langle O(\phi) \rangle_S. \tag{2.6}$$

The above statement can be proved for the real action. On the other hand no general theorem has been found, which states clearly what happens whenever the action S becomes complex.

2.2.1 Real Langevin Equation

In the above section, we saw that the Langevin equation describes a relaxation dynamics, which has an asymptotic property given in Eq. (2.6). This subsection is devoted to give an overview of the proof of that statement [3].

Consider a system with one degree of freedom say, x. The formalism can be generalized to higher degrees of freedom as well. The Langevin equation for such a system can be written as

$$\frac{dx}{dt} = S'(x) + \eta(t). \tag{2.7}$$

The solution $x_\eta(t)$ of Eq. (2.7) subjected to a Gaussian white noise has a probability distribution at a particular time, $P(\phi, t)$. The expectation value of any quantity $O(x_n)$ with respect to white noise can be written as

$$\langle O(x_n(t)) \rangle_\eta = \int dx \; P(x;t) O(x). \tag{2.8}$$

Taking derivative with respect to fictitious time t on both sides of the above equation we get

$$\frac{d\langle O(x_n(t))\rangle_\eta}{dt} = \int dx\ O(x)\delta_t P(x;t). \tag{2.9}$$

Rewriting Langevin equation (2.7) as

$$dx = -S'(x)dt + dW(t) \tag{2.10}$$

with

$$dW(t) = \int_t^{t+dt} d\tau\ \eta(\tau) \tag{2.11}$$

and by using Eq. (2.5) one can obtain

$$\langle dW(t)^2\rangle_\eta = \int_t^{t+dt} d\tau \int_t^{t+dt} d\tau'\ \langle \eta(\tau)\eta(\tau')\rangle_\eta = 2dt. \tag{2.12}$$

For calculating $d\langle O(x_n(t))\rangle_\eta$ up to first order in dt, we have to expand $dO(x_n(t))$ up to second order in dx.

$$\begin{aligned}\frac{d\langle O(x_n(t))\rangle_\eta}{dt} &= \langle -O'(x)S'(x) + O''(x)\rangle_\eta \\ &= \int dx\Big[-O'(x)S'(x) + O''(x)\Big]P(x,t) \\ &= dxO(x)[\delta_x S'(x) + \delta_x^2]P(x;t). \end{aligned} \tag{2.13}$$

After doing integration by parts and assuming zero boundary conditions at infinity we can combine Eq. (2.9) and Eq. (2.13). This gives us the Fokker-Planck (FP) equation

$$\delta_t P(x;t) = -H_{FP}P(x;t) \tag{2.14}$$

with the Fokker-Planck Hamiltonian

$$H_{FP} = -\delta_x(\delta_x + S'(x)). \tag{2.15}$$

The stationary point of Eq. (2.15) is easily found to be $P(x) \sim e^{-S(x)}$. Moreover, we can rewrite Eq. (2.15) upon the transformation

$$P(x,t) = \psi(x,t)e^{\frac{S(x)}{2}} \tag{2.16}$$

into the Schrödinger like equation

$$\dot{\psi}(x,t) = -H_{FP}\psi(x,t) \tag{2.17}$$

where

$$H_{FP} = \left(-\delta_x + \frac{S'(x)}{2}\right)\left(\delta_x + \frac{S'(x)}{2}\right). \tag{2.18}$$

The operator Eq. (2.18) is self-adjoint and, if $|\lim_{x\to\infty} S'(x) \to \infty|$, the spectrum of its eigenvalues is non-negative and discrete

$$H\psi_n = E_n\psi_n \tag{2.19}$$

with the ground state $\psi_0 \sim e^{-S(x)/2}$ annihilating Eq. (2.18). Therefore we can rewrite $\psi(x,t)$ on the base of the eigenvectors of H_{FP}

$$\psi(x,t) = c_0 e^{-\frac{S(x)}{2}} + \sum_n c_n\psi_n(x)e^{-E_n t} \to c_0 e^{-\frac{S(x)}{2}} \tag{2.20}$$

and the correct distribution $P(x) \sim e^{-S(x)}$ is reached exponentially fast.

2.2.2 Complex Extension

In this section we shall see how Langevin dynamics can be extended to the case of complex actions $S(x)$. A straightforward adaptation of Eq. (2.7) for a complex action is still possible and would lead to the Langevin equation

$$\dot{x} = -S'(x) + \eta(t). \tag{2.21}$$

Now, the FP equation (2.14) is expected to have a complex weight

$$P(x) \sim e^{-S(x)} \tag{2.22}$$

as a stationary solution. However, being complex valued $P(x)$ cannot be considered as probability distribution function as in Eq. (2.8). Moreover, the FP Hamiltonian is no longer a self-adjoint operator, so that a proof of exponentially fast convergence to the unique solution cannot be provided. The way to proceed then is to write the Langevin equation (2.21) using two real variables x and y with $z = x + iy$[3].

$$\dot{x} = K_x + \eta_R, \tag{2.23}$$
$$\dot{y} = K_y + \eta_I, \tag{2.24}$$

where the $K_x = -\mathrm{Re}(\delta_z S(z))$ and $K_y = -\mathrm{Im}(\delta_z S(z))$ are the two drift terms.

The noise correlators corresponding to complex noise $\eta = \eta_R + \eta_I$ can be derived from Eq. (2.5).

$$\langle \eta_R(t)\eta_R(t')\rangle = 2\delta(t-t'), \tag{2.25}$$
$$\langle \eta_I(t)\eta_I(t')\rangle = 2\delta(t-t'), \tag{2.26}$$
$$\langle \eta_R(t)\eta_I(t')\rangle = 0. \tag{2.27}$$

For the complex Langevin (CL), we will have the real probability distribution, $P(x; y; t)$ as a function of x and y, which satisfies the following FP equation

$$\delta_t P(x, y; t) = -H_{FP} P(x, y; t), \tag{2.28}$$
$$H_{FP} = -\delta_x^2 - \delta_x \mathrm{Re}(S') - \delta_y \mathrm{Im}(S'). \tag{2.29}$$

Now the expectation of an observable can be defined as

$$\langle O(z)\rangle_\eta = \int dxdy \; P(x, y, t) O(x + iy). \tag{2.30}$$

In case of complex Langevin we cannot do the analogous analysis like we did in the previous section of FP Hamiltonian due to the complexity of the action. The action does not guarantee the positive semi-definiteness of H_{FP}, which was crucial to prove the asymptotic property of the probability distribution. Furthermore, unstable solutions of Eq. (2.23) can be found in the complex plane and that was believed to inevitably spoil the dynamics when solved numerically. These two problems have been more recently addressed, allowing CL to become one of the most acknowledged methods when it comes to system affected by the sign problem.

The issue of instabilities on the lattice was the first to be successfully and consistently solved. The discrete analogue of complex Langevin equations for a field ϕ are

$$\phi_x^R(n+1) = \phi_x^R(n) - \epsilon \; \mathrm{Re}(S'(x_n, y_n)) + \sqrt{\epsilon}\eta_x^R(n), \tag{2.31}$$
$$\phi_x^I(n+1) = \phi_x^I(n) - \epsilon \; \mathrm{Im}(S'(x_n, y_n)) + \sqrt{\epsilon}\eta_x^I(n), \tag{2.32}$$

where ϵ is the discrete Langevin time step and x labels the sites of the lattice. When the system is brought near an unstable trajectory $\tau(\phi^R, \phi^I)$, the drifts can potentially lead the fields to infinity, in a finite Langevin time $t_L = \epsilon n$. It turns out that careful integration in the form of adaptive step size on along those trajectories is enough to completely remove the problem.

3 Non-Lattice Method

In this section, we will apply the complex Langevin method to the supersymmetric anharmonic oscillator using the non-lattice Fourier-expansion method as discussed in [4].

The action for the model is

$$S = \int_0^\beta dt \left[\frac{1}{2}(\partial_t \phi)^2 + h'(\phi)^2 + \bar{\psi}(\partial_t + h''(\phi))\Psi\right]. \tag{3.1}$$

Here ϕ is a real scalar field and ψ is a one component Dirac field. Both the fields satisfy periodic boundary conditions

$$\phi(t+\beta) = \phi(t) \quad \text{and} \quad \psi(t+\beta) = \psi(t). \tag{3.2}$$

Let us use the Fourier expansion of the fields

$$\phi(t) = \sum_{n=-\Lambda}^{\Lambda} \tilde{\phi}_n e^{in\omega t} \quad \text{and} \quad \psi(t) = \sum_{n=-\Lambda}^{\Lambda} \tilde{\psi}_n e^{in\omega t}, \tag{3.3}$$

where $\omega = 2\pi/\beta$ with n an integer and Λ the UV cutoff. The reality conditions $\phi^*(t) = \phi(t)$ gives $\phi_n^* = \tilde{\phi}_{-n}$ and similarly for fermionic field $\psi_n^* = \tilde{\psi}_{-n}$. We make use of the integral

$$\int_0^\beta e^{in\omega t} dt = \beta\delta_{n0}. \tag{3.4}$$

Integrating out the fermionic part of the action, we obtain an effective action

$$S_{\text{eff}} = S_B - \ln\det M. \tag{3.5}$$

Here, M is a $(2\Lambda+1)\times(2\Lambda+1)$ matrix. The row and column index $(2\Lambda+1)$ represents the number of modes or the number of degrees of freedom. The exact form of the fermionic matrix M depends on the potential $h(\phi)$.

The partition function takes the form

$$\begin{aligned} Z &= \int_{\text{PBC}} \mathcal{D}\phi\mathcal{D}\psi \; e^{-S(\phi,\psi)} \\ &= \int_{\text{PBC}} \mathcal{D}\phi \; e^{-S_B} \det M \\ &= \int_{\text{PBC}} \mathcal{D}\phi \; e^{-S_{\text{eff}}}. \end{aligned} \tag{3.6}$$

3.1 Non-Lattice Complex Langevin

We solve the following discretized version of Langevin equation as per Sec. 2.2.2. Here l is the Langevin time; it is a fictitious parameter, not the real time dimension.

$$\phi_n(l+\epsilon) = \phi_n(l) - \epsilon\frac{\delta S_{\text{eff}}}{\delta\phi_{-n}} + \sqrt{\epsilon}\eta_n(l). \tag{3.7}$$

Eq. (3.7) is the evolution equation followed by each Fourier mode. The white noise function $\eta_n(l)$ in the evolution equation has a Gaussian distribution

$$\propto \exp\left(\frac{-1}{4}\sum_l \sum_{-\Lambda}^{\Lambda} |\eta_n(l)|^2\right) = \exp\left[\frac{-1}{4}\sum_l \left(\eta_0^2(l) + \sum_1^{\Lambda} |2\eta_n(l)|^2\right)\right], \tag{3.8}$$

where, $\eta_0(l) = \sqrt{2}x_0(l)$ and $\eta_{\pm n}(l) = x_n(l) \pm iy_n(l)$. Both $x_n(l)$ and $y_n(l)$ obey the standard normal distribution $N(0,1)$, independently.

3.2 Anharmonic Oscillator

The anharmonic oscillator potential is given by

$$h(\phi) = \frac{1}{2}m\phi^2 + \frac{1}{4}g\phi^4, \tag{3.9}$$

$$h'(\phi) = m\phi + g\phi^3. \tag{3.10}$$

For the potential Eq. (3.9), the action is

$$S = \int_0^\beta \left(\frac{1}{2}(\partial_t\phi)^2 + \frac{1}{2}m^2\phi^2 + mg\phi^4 + \frac{1}{2}g^2\phi^6 + \bar{\psi}(\partial_t + m + 3g\phi^2)\psi \right) dt. \tag{3.11}$$

In terms of the Fourier modes, the action can be written as $S = S_B + S_F$, where

$$S_B = \beta \left[\frac{1}{2} \sum_{n=-\Lambda}^{\Lambda} (n^2\omega^2 + m^2)\tilde{\phi}_n\tilde{\phi}_{-n} + mg(\tilde{\phi}^4)_0 + \frac{1}{2}g^2(\tilde{\phi}^6)_0 \right] \tag{3.12}$$

and

$$S_F = \sum_{n,m=-\Lambda}^{\Lambda} \tilde{\bar{\psi}}_n \beta \left((in\omega + m)\delta_{nm} + 3g \sum_{l=-2\Lambda}^{2\Lambda} (\tilde{\phi}^2)_l \delta_{n,m+l} \right) \psi_m, \tag{3.13}$$

where

$$M_{n,m} = \beta \left\{ (in\omega + m)\delta_{nm} + 3g \sum_{l=-2\Lambda}^{2\Lambda} (\tilde{\phi}^2)_l \delta_{n,m+l} \right\}. \tag{3.14}$$

The gradient of the action is

$$\begin{aligned} \frac{\partial S_{\text{eff}}}{\partial \tilde{\phi}_n} &= \beta \left[(n^2\omega^2 + m^2)\tilde{\phi}_{-n} + 4mg(\tilde{\phi}^3)_{-n} + 3g^2(\tilde{\phi}^5)_{-n} \right] \\ &\quad - \text{Tr}\left(\frac{\partial M}{\partial \tilde{\phi}_n} M^{-1} \right). \end{aligned} \tag{3.15}$$

Here, "Tr" represents the trace of the $\left(\frac{\partial M}{\partial \tilde{\phi}_n} M^{-1}\right)_{(2\Lambda+1)\times(2\Lambda+1)}$ matrix.

3.2.1 Extraction of Masses

Here we are discussing the formalism for the bosonic mass gaps. The formalism remains same for fermionic part as well.

We extract masses from the exponential decay of the two point correlators. We can calculate the two point correlation function as

$$\begin{aligned} G_B(t) &= \langle \phi(0)\phi(t) \rangle \\ &= \left\langle \sum_{m,n=-\Lambda}^{\Lambda} \tilde{\phi}_m\tilde{\phi}_n e^{i\omega nt} \right\rangle \qquad (3.16) \\ &= \langle \tilde{\phi}_0^2 \rangle + \sum_{n=1}^{\Lambda} \langle \tilde{\phi}_n\tilde{\phi}_{-n} \rangle (e^{iq\omega t} + e^{-in\omega t}) + \sum_{m+n\neq 0} \langle \tilde{\phi}_m\tilde{\phi}_n \rangle e^{in\omega t} \\ &= c_0 e^{-m_B t} + \cdots \qquad (3.17) \\ &= b_0 + \sum_{n=1}^{\Lambda} 2b_n \cos(n\omega t), \qquad (3.18) \end{aligned}$$

where $b_0 = \langle \tilde{\phi}_0^2 \rangle$ and $b_n = \langle \tilde{\phi}_n\tilde{\phi}_{-n} \rangle$. In sufficiently large time the mass is obtained as a plateau. The mass can be extracted out from Eq. (3.16) as

$$-\ln \frac{G_B(t+\Delta t)}{G_B(t)} = -\ln \frac{c_0 e^{-m_b(t+\Delta t)}}{c_0 e^{-m_b t}} = -\ln e^{-m_b(\Delta t)} = m_B(\Delta t). \tag{3.19}$$

Eq. (3.19) can also be manipulated as

$$m_B = -\frac{1}{\Delta t} \ln \frac{G_B(t+\Delta t)}{G_B(t)} = -\frac{1}{\Delta t}(\ln G_B(t+\Delta t) - \ln G_B(t)). \tag{3.20}$$

For sufficiently large time t the finite difference becomes derivative and we can write the Eq. (3.20) as

$$m_B = -\frac{d}{dt} \ln G_B(t) = \frac{\sum_{n=1}^{\Lambda} 2n\omega b_n \sin(n\omega t)}{b_0 + \sum_{n=1}^{\Lambda} 2b_n \cos(n\omega t)}. \tag{3.21}$$

The function (3.16) oscillates with respect to t with the order of the cutoff. This is nothing but the Gibbs phenomenon due to the sharp cutoff in the sum over Fourier modes. To avoid this problem, we extend the sum over n up to some Λ_0. For $\Lambda < n \leq \Lambda_0$, we define b_n as $b_n \sim \frac{d_1}{n^2} + \frac{d_2}{n^4}$ [4]. The coefficients d_1 and d_2 can be calculated from the results at $n = \Lambda - 1, \Lambda$, that is,

$$\begin{aligned} b_{\Lambda-1} &= \frac{d_1}{(\Lambda-1)^2} + \frac{d_2}{(\Lambda-1)^4} \\ b_\Lambda &= \frac{d_1}{(\Lambda)^2} + \frac{d_2}{(\Lambda)^4} \end{aligned} \tag{3.22}$$

$$\begin{aligned} \implies d_1 &= \frac{\Lambda^4 b_\Lambda - (\Lambda-1)^4 b_{\Lambda-1}}{2\Lambda - 1} \\ d_2 &= \frac{(\Lambda-1)^4\Lambda^2 b_{\Lambda-1} - (\Lambda-1)^2\Lambda^4 b_\Lambda}{2\Lambda-1}. \end{aligned} \tag{3.23}$$

In short, we can extend Eq. (3.16) to the sum

$$G_B(t) \to b_0 + \sum_{n=1}^{\Lambda} \langle \tilde{\phi}_n \tilde{\phi}_{-n} \rangle \cos(\omega n t) + \sum_{n=\Lambda+1}^{\Lambda_0} \left(\frac{d_1}{n^2} + \frac{d_2}{n^4} \right) \cos(\omega n t), \tag{3.24}$$

where d_1 and d_2 come from Eq. (3.24). We can do analogous analysis for $\mathrm{Re}(c_n)$ for extracting the fermionic mass.

4 Simulation Details and Results

In this section we present our simulation results.

4.1 Mass Gap

We simulated the model for physical parameters $m_{\text{phys}} = 10$, $g_{\text{phys}} = 0$ and $\beta = 1$. Simulations were performed for $\Lambda = 4, 8, 12, 16, 32$ values. We used adaptive Langevin step size ($\Delta\tau \leq 5 \times 10^{-3}$), thermalization steps $N_{\text{therm}} = 10^4$, and generation steps $N_{\text{gen}} = 10^5$. Measurements were taken with a gap of 10 steps.

Fig. (2) shows bosonic mass versus physical time. The plot is based on the data in the Tab. 1. The mass gaps are coming out as plateaus. The mass gaps for each Λ are the same because we are considering only the free case. The mass gaps will depend on Λ for finite coupling or full action.

4.2 Bosonic Action

The bosonic action can also be used to see SUSY breaking. SUSY is preserved for the model given in Eq. (3.1) with supersymmetric anharmonic potential Eq. (3.9) [5].

It has been studied that if SUSY is preserved, the expectation value of the bosonic action simply counts the number of degrees of freedom [6]. In our case,

$\Lambda = 4$	$\Lambda = 8$	$\Lambda = 12$	$\Lambda = 16$	$\Lambda = 32$
$b_0 = 0.105$	$b_0 = 0.101000$	$b_0 = 0.101284$	$b_0 = 0.113081$	$b_0 = 0.122207$
$b_1 = 0.067$	$b_1 = 0.073000$	$b_1 = 0.0742187$	$b_1 = 0.075271$	$b_1 = 0.077000$
$b_2 = 0.041$	$b_2 = 0.038481$	$b_2 = 0.038121$	$b_2 = 0.035918$	$b_2 = 0.038821$
$b_3 = 0.022$	$b_3 = 0.022550$	$b_3 = 0.022063$	$b_3 = 0.024542$	$b_3 = 0.028266$
$b_4 = 0.014$	$b_4 = 0.013550$	$b_4 = 0.013422$	$b_4 = 0.013629$	$b_4 = 0.013333$
0	$b_5 = 0.009200$	$b_5 = 0.009371$	$b_5 = 0.008956$	$b_5 = 0.009217$
0	$b_6 = 0.006000$	$b_6 = 0.006586$	$b_6 = 0.006431$	$b_6 = 0.006592$
0	$b_7 = 0.005000$	$b_7 = 0.004966$	$b_7 = 0.004930$	$b_7 = 0.005137$
0	$b_8 = 0.003900$	$b_8 = 0.003890$	$b_8 = 0.003869$	$b_8 = 0.003942$
0	0	$b_9 = 0.003080$	$b_9 = 0.002939$	$b_9 = 0.003080$
0	0	$b_{10} = 0.002622$	$b_{10} = 0.002467$	$b_{10} = 0.002448$
0	0	$b_{11} = 0.002132$	$b_{11} = 0.002037$	$b_{11} = 0.002099$
0	0	$b_{12} = 0.001809$	$b_{12} = 0.001778$	$b_{12} = 0.001714$
0	0	0	$b_{13} = 0.001558$	$b_{13} = 0.001472$
0	0	0	$b_{14} = 0.001345$	$b_{14} = 0.001283$
0	0	0	$b_{15} = 0.001168$	$b_{15} = 0.001104$
0	0	0	$b_{16} = 0.001038$	$b_{16} = 0.000997$
0	0	0	0	$b_{17} = 0.000881$
0	0	0	0	$b_{18} = 0.000771$
0	0	0	0	$b_{19} = 0.000691$
0	0	0	0	$b_{20} = 0.000633$
0	0	0	0	$b_{21} = 0.000591$
0	0	0	0	$b_{22} = 0.000519$
0	0	0	0	$b_{23} = 0.000487$
0	0	0	0	$b_{24} = 0.000461$
0	0	0	0	$b_{25} = 0.000417$
0	0	0	0	$b_{26} = 0.000384$
0	0	0	0	$b_{27} = 0.000363$
0	0	0	0	$b_{28} = 0.000334$
0	0	0	0	$b_{29} = 0.000311$
0	0	0	0	$b_{30} = 0.000300$
0	0	0	0	$b_{31} = 0.000272$
0	0	0	0	$b_{32} = 0.000264$
$d_1 = 0.25$	$d_1 = 0.36$	$d_1 = 0.273903$	$d_1 = 0.284568$	$d_1 = 0.408735$
$d_2 = -0.6$	$d_2 = -8.8$	$d_2 = -1.92787$	$d_2 = -4.90473$	$d_2 = -141.388$

Table 1: The coefficients b_n, d_1, and d_2 obtained from simulations for $\Lambda = 4, 8, 12, 16$, and 32.

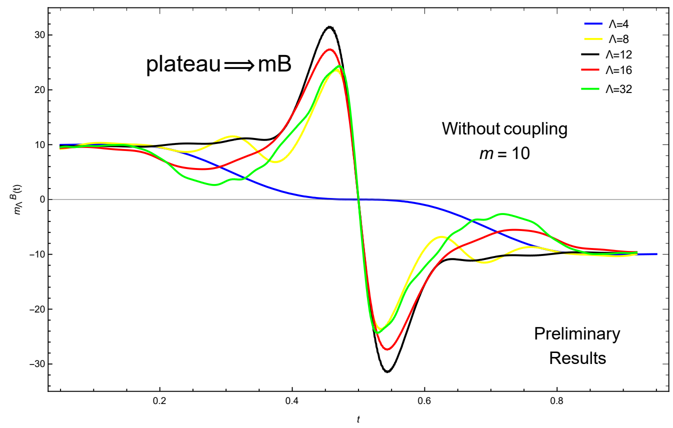


Figure 2: Bosonic physical mass for SUSY anharmonic oscillator with physical parameter $m_{\text{phys}} = 10$ and $g_{\text{phys}} = 0$ versus physical time t.

it is expected that $\langle S \rangle = 2\Lambda + 1$ and $\langle S^B \rangle = \frac{2\Lambda+1}{2}$ where $(2\Lambda + 1)$ are the number of modes or the number of degree of freedom. Thus we have

$$\langle S_B \rangle = \begin{cases} \neq \frac{2\Lambda+1}{2} & \text{SUSY broken,} \\ = \frac{2\Lambda+1}{2} & \text{SUSY unbroken.} \end{cases} \tag{4.1}$$

Data given in Tab. 2 are generated from simulations. The simulation parameters were the same as used for obtaining the mass gaps. The bosonic action expectation values for all Λ are close to the corresponding theoretical values.

Λ	$\langle S^B \rangle$
4	4.4737 ± 2.0583
8	8.56261 ± 2.9401
12	12.6656 ± 3.4858
16	16.1544 ± 3.9966
32	32.5708 ± 5.9173

Table 2: The expectation values of the bosonic action S^B for SUSY anharmonic oscillator with physical parameters $m_{\text{phys}} = 10$ and $g_{\text{phys}} = 0.0$. The simulations were performed for $\Lambda = 4, 8, 12, 16$, and 32.

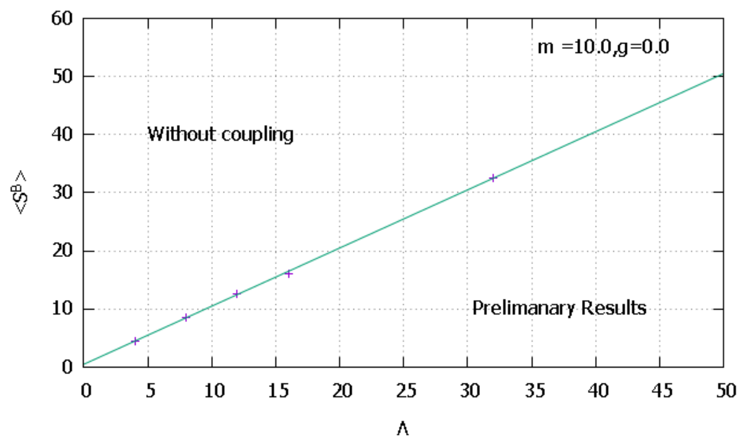
Fig. 3(a) is based on Tab. 2. The data points corresponding to different Λ values fall over the theoretical line. Fig. 3(b) shows the Langevin evolution history and Λ dependence of the bosonic action.The average value of the bosonic action $\langle S_B \rangle = \frac{2\Lambda+1}{2}$, and it was independent of the physical parameters m_{phy} and g_{phy}.

We can conclude from Tab. 2 and Fig. 3 that SUSY is preserved for the SUSY anharmonic oscillator potential.

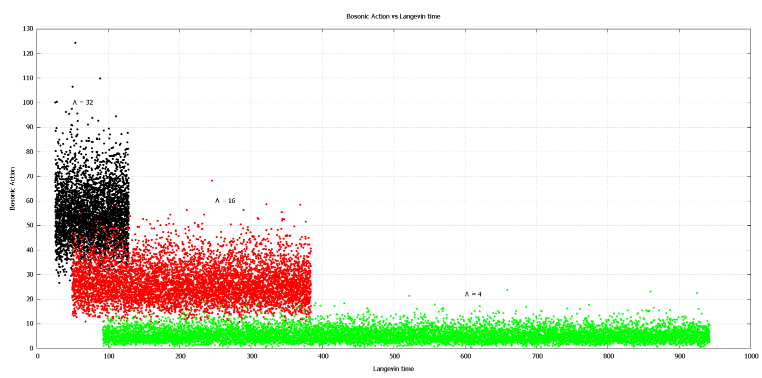
5 Conclusions and Future Directions

We looked at the theoretical aspects of the complex Langevin dynamics and saw its application in taming the sign problem. Later, we introduced a new simulation method (non-lattice formalism), which enables non-perturbative studies of supersymmetric gauge theories in one dimension. We probed the SUSY breaking by looking at the mass gaps and bosonic degrees of freedom.

Work is in progress for calculating the fermionic mass from the fermionic correlator. Our next target is to show that supersymmetry is preserved for the complete supersymmetric model. We will also apply this method for several types of one-dimensional gauge theories.



(a)



(b)

Figure 3: (a) Expectation value of the bosonic action with Λ for parameters $m = 10.0$ and $g = 0.0$. The line is the analytical result, while the points are computed using complex Langevin dynamics. (b) The bosonic action against the Langevin time for $\Lambda = 4, 16$, and 32.

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